

ECM3401 Project Logbook

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Log of Activities

Date (W/C)	Key Activities	Outcomes	Comments / Action Plans
12/01/2026	Create script to take in IMU data and turn it into a readable format based on dummy data recorded personally. Work on ethics application.	Ethics form submitted (Sunday 18th Jan).	Ethics application not submitted earlier due to illness and technical issues with access to Worktribe.
19/01/2026	Continue script creation, make forms for participant application and Reddit account.	Ethics approval returned for modification (Friday 23rd Jan). Script created for preparation of data.	Waiting on ethics approval.
26/01/2026	Create script for preprocessing of data.	Script partially made, reading in and further slight cleaning.	Contemplation regarding the application of forces and what additional features to extract from them.
02/02/2026	Continue preprocessing script, implement rolling average.	Ethics application resubmitted (Friday 6th Feb).	

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		Rolling average implemented.	
09/02/2026	Implement final touches to preprocessing.	Application returned (Monday 9th Feb). Changes made and resubmitted. Application passed on for further review (Monday 9th Feb).	
16/02/2026	Implement window creation of preprocessed data.	Created partial method for generating windows.	Implemented method is not automated yet and needs to be adapted for more files.
23/02/2026	Further work on window creation.	Ethics review status changed to "up for review" (Monday 23rd Feb). Finished implementation of window creation.	
02/03/2026	Lay out/work on information extraction for windows.	Basic statistical values extracted from windows	
09/03/2026	Enquire as to status of ethics application (Monday 9th March). Slight work on window extraction.	Still waiting for ethics Have cleaned up window extraction section and added labels to windows	
16/03/2026	Modify all stages of pipeline (accounting for	Application returned for changes (Monday	Example data was used to create the general pipeline

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	change in metrics and formatting).	16th March). Changes made (Tuesday 17th). Decided not to go forward with data collection due to time. Project properly pivoting to the new dataset (SimRa) on 19th March.	and preprocessing stages before the pivot. Thankfully the SimRa data still contains very similar metrics, so pivoting is not difficult.
23/03/2026	Convert pipeline to use new format of data. Submission of poster for presentation.	Functional single-stage model achieved (23rd March). Poster submitted for printing.	Critical feedback from supervisor received; changes implemented. Realised model could be improved by separating detection and characterisation.
30/03/2026	Single stage pipeline clearly struggling with large amount of data; separate machine learning model into two: one for detection and another for characterisation.	Managed to create baseline framework, but the second model is struggling due to the size of the dataset.	Need to somehow trim down dataset?
06/04/2026	Implement copy of first stage into a new middle stage due to 2nd stage difficulties.	Created stage 2 that acts as a sifter; largely lowers imbalance	
13/04/2026	Further development of 3-stage model.	Functional three-stage pipeline achieved (16th April).	Not the final version as of 16/04; needed to add in some more methods to save metrics.

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20/04/2026	Implement further saving of metrics for reference within the final report.	Metric saving implemented.	Was stupid not to do this earlier, as it meant that any time I wanted metrics previously I relied solely on console outputs.
27/04/2026	Submission of project.	Final report submitted.	Logbook/data/code not submitted yet due to sleep deprivation.